HyperMamba: A Spectral–Spatial Adaptive Mamba for Hyperspectral Image Classification

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***Abstract*— Transformers have significantly advanced hyperspectral image (HSI) classification through their proficiency in modeling long sequences. However, the high dimensionality of HSIs poses a particular challenge for Transformers due to their quadratic computational complexity. In natural language processing, state-space models (SSMs) such as Mamba hold great promise for handling long sequence tasks with significantly reduced computational overhead. However, the original Mamba lacks consideration for the spectral and spatial information inherent in HSIs. Inspired by this, we propose the HyperMamba, a novel spectral–spatial adaptive Mamba for HSI classification. The core idea of HyperMamba involves adaptively scanning spatial neighborhood pixels and dynamically enhancing spectral bands for spectral scanning based on acquired spatial neighborhood information. Specifically, HyperMamba consists of two core modules: the spatial neighborhood adaptive scanning (SNAS) module and the spectral adaptive enhancement scanning (SAES) module. Initially, the SNAS module analyzes the spectral characteristics of classified pixels, adaptively selecting the optimal neighborhood for spatial scanning by balancing spatial neighborhood information and local spatial structure. Subsequently, the SAES module dynamically enhances the spectral features of classified pixels using neighborhood spectral information and conducts spectral scanning. Finally, the spectral features of the target pixels are fed into a single fully connected layer classifier, achieving high-precision HSI classification. Extensive experiments demonstrate the effectiveness of HyperMamba, surpassing state-of-the-art methods across three widely used HSI datasets. The code will be available at https://github.com/chiangliu/HyperMamba.**

***Index Terms*— Deep neural network, hyperspectral image (HSI) classification, Mamba.**

I. Introduction

HYPERSPECTRAL imaging is a crucial remote sensing (RS) technique that provides high-resolution spectral data of the Earth's surface. Hyperspectral images (HSIs) contain rich spatial and spectral information at the same time, each pixel representing the spectral reflectance of different wavelengths [1]. Due to its ability to display subtle spectral changes between different objects, HSIs play an important role in Earth observation tasks such as mineral exploration [2], environmental monitoring [3], and precision agriculture [4].

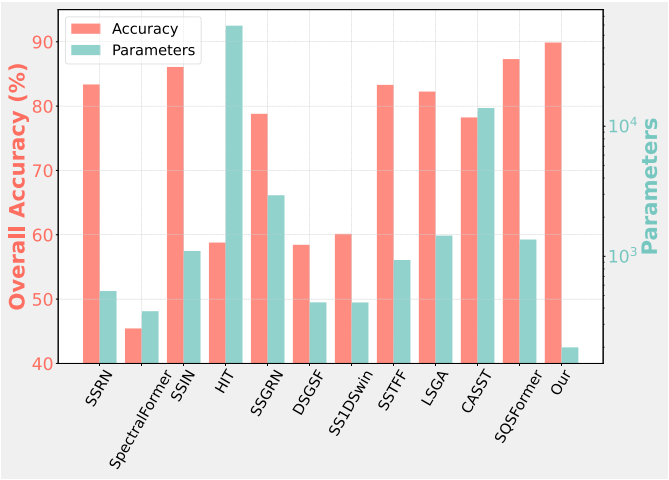


Fig. 1. Comparison of accuracy and parameter quantity among various algorithms on the IP dataset with ten training samples per class.

HSI classification has received widespread attention in the RS field, which aims to accurately assign each pixel to its corresponding land-cover class [5]. Early studies primarily focused on exploring the spectral characteristics of HSIs for classification. Traditional methods such as neural networks [6], support vector machines [7], and multinomial logistic regression [8] have been widely applied to pixel-wise HSI classification tasks. Due to the challenges of spectral redundancy and noise in HSIs [9], [10], many methods focus on effective feature extraction or dimensionality reduction using techniques such as principal component analysis (PCA) [11], independent component analysis (ICA) [12], and linear discriminant analysis (LDA) [13]. In addition, some researchers have integrated spatial information into pixel classifiers, and some spectral–spatial features-based classification frameworks have been developed, such as extended morphological profile [14], directional morphological profile [15], and edge-preserving filtering [16]. Although these methods have shown good performance, they heavily rely on manual design or shallow descriptors and struggle to handle complex environments.

In the past few years, deep learning has achieved significant success in tasks like image classification [17], segmentation, and object detection due to its strong feature extraction capabilities [18], [19], [20], [21]. To leverage spatial–spectral features in HSIs, researchers have developed various deep learning-based HSI classification models [22], [23]. Currently,

Received 14 May 2024; revised 30 September 2024; accepted 14 October 2024. Date of publication 17 October 2024; date of current version 7 November 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62425109, Grant U22B2014, and Grant 62221002; and in part by the Science and Technology Plan Project Fund of Hunan Province under Grant 2022RC3064. *(Corresponding author: Leyuan Fang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3482473

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the mainstream deep learning models for HSI classification can be broadly categorized into three types: convolutional neural networks (CNNs) [24], graph convolutional neural networks (GCNs) [25], and Transformers [26]. For CNN-based methods, Roy et al. [27] proposed a hierarchical 3-D–2-D CNN model that combines 2-D and 3-D convolutional kernels for HSI classification. Wei et al. [28] first encoded the Extended Morphological Attribute Profile into quaternions, creating a compact yet rich representation called EQR. They then embedded EQR into a quaternion convolutional neural network (QCNN) to propose a novel dual-branch EQR-QCNN model for multisource RS data classification. Zhou et al. [29] developed a separable quaternion convolutional neural network (SQNet) for HSI classification, incorporating a novel quaternion feature encoder and key modules like separable quaternion convolutions and attention mechanisms, which resulted in high-accuracy classification results. Li et al. [30] proposed a new dual-channel spatial, spectral, and multiscale attention convolutional long short-term memory (LSTM) neural network that effectively utilizes the complementarity between HSIs and light detection and ranging (LiDAR) data. With respect to GCNs, Yang et al. [31] proposed a multifrequency graph convolutional network with cross-modality mutual enhancement (MFGCN-CME), effectively capturing and adaptively aggregating low- and high-frequency multiscale features from HSI and LiDAR data. Wang et al. [25] proposed an end-to-end spectral–spatial graph reasoning network (SSGRN) that generates dynamic graph structures through flexible homogeneous areas. This network can adaptively capture contexts across different objects or channels, even when they are irregularly distributed. For Transformer-based architectures, Hong et al. [32] revisited HSI classification from a sequential perspective and proposed SpectralFormer, which learns spectrally local sequence information from neighboring bands. Peng et al. [33] proposed a spatial–spectral Transformer with cross-attention (CASST), consisting of two branches: one captures fine spatial details of HSIs, while the other extracts spectral features and models their interdependencies. Yang et al. [34] presented interactive transformer and CNN with a multilevel feature fusion network (ITCNet), which integrates Transformers and CNNs and employs a multilevel feature fusion network for HSI classification. Despite the advances in HSI classification mentioned above, several limitations still exist. CNN-based methods are constrained by their local receptive fields, hindering the capture and learning of complex spatial–spectral feature representations. GCN-based methods incur high computational costs, particularly for large graphs, often failing to classify or identify materials in extensive hyperspectral scenes [35]. Although Transformer-based methods can learn long-range dependencies, effectively extracting long-sequence spatial and spectral information from high-dimensional HSIs remains challenging due to computational constraints and quadratic complexity.

Recently, Mamba presented a promising solution for handling long sequences [36]. By integrating a selective scanning mechanism and hardware-aware algorithm into the state-space model (SSM) [37], Mamba can establish long-distance dependencies while maintaining linear computational complexity, showcasing excellent performance in long sequence language modeling tasks. Consequently, several studies have initiated exploration into employing the Mamba model for computer vision tasks [38], [39], [40], exemplified by Vim [41] and VMamba [42]. These methods entail scanning the spatial domain from multiple directions, converting 2-D images into 1-D sequences for SSM-based processing and achieving exceptional performance in tasks like image classification and segmentation.

By converting the original single-direction scanning into multidirectional scanning, the Mamba variants have achieved significant success in natural image processing. However, since Mamba was originally designed for 1-D data, merely modifying the scanning direction does not fully capture the spatial and spectral information inherent in high-dimensional HSIs. In this article, we propose HyperMamba, a purely Mamba-based model tailored for HSI classification. The core idea of HyperMamba involves adaptive scanning of spatial pixels and dynamically enhancing spectral bands for spectral scanning based on acquired spatial neighborhood information. Specifically, we first design a spatial neighborhood adaptive scanning (SNAS) module, which analyzes the spectral characteristics of classified pixels, adaptively selecting the optimal neighborhood for spatial scanning by balancing the amount of spatial neighborhood information and local spatial structure. Next, we propose a spectral adaptive enhancement scanning (SAES) module, which dynamically enhances the spectral features of classified pixels using neighborhood spectral information and conducts spectral scanning. Finally, the spectral features of the target pixels are fed into a simple classifier to achieve high-precision HSI classification. Our main contributions are summarized as follows.

1) We analyze the limitations of Mamba in processing HSIs, as it disregards spectral and spatial characteristics. To address this, we propose the HyperMamba to enhance spatial–spectral interactions in Mamba for HSI classification.
2) To address the issues of spatial disruption caused by large patches and context loss from small patches when Mamba converts 2-D blocks into 1-D sequences, we propose the SNAS module to adaptively select the optimal neighborhood for spatial scanning by balancing neighborhood information and local structure.
3) To address the interference between spectral and spatial information that arises during the scanning process when Mamba converts high-dimensional HSI data into 1-D sequences, we propose the SAES module, which dynamically enhances the spectral features of classified pixels using neighborhood spectral information.
4) Comprehensive experiments conducted on three widely recognized benchmark datasets for HSI classification, namely Indian Pines (IP), University of Houston, and WHU-Hi-HongHu (WHHH), demonstrate that our method achieves state-of-the-art performance, as shown in Fig. 1.

II. Preliminaries

A. State-Space Models

SSMs [43] are a type of sequence model, commonly employed to model continuous linear time-invariant (LTI)

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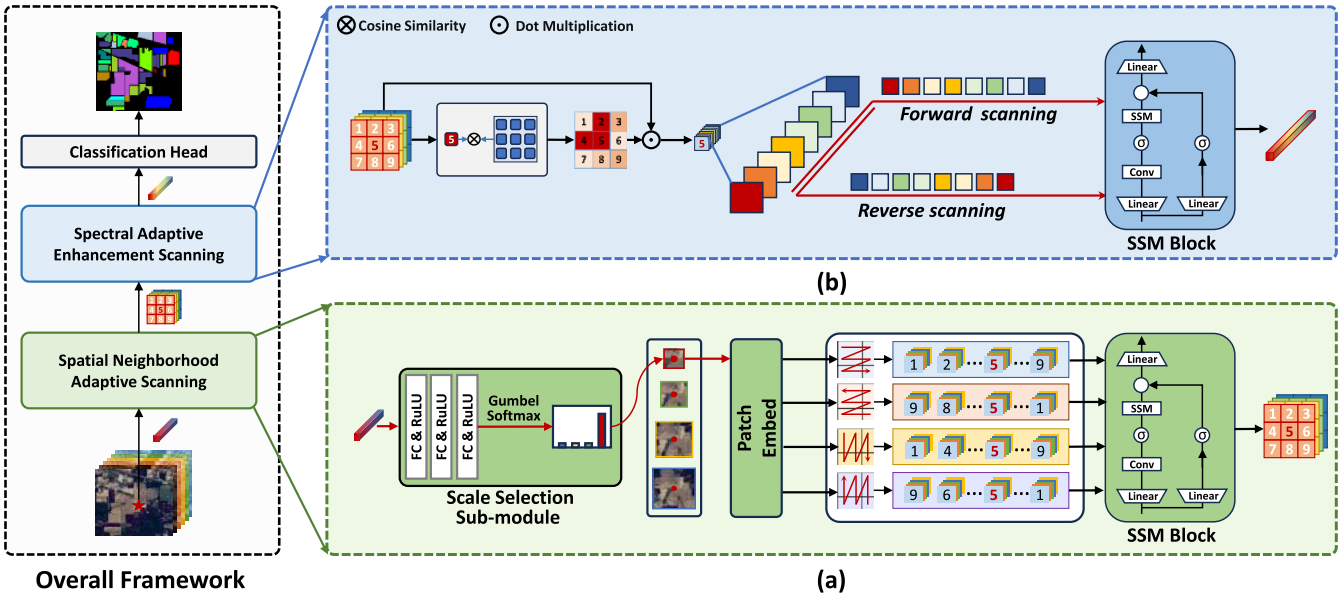


Fig. 2. Overview of the HyperMamba framework, which comprises two core modules: (a) SNAS module and (b) SAES module. Initially, the SNAS module analyzes the spectral characteristics of classified pixels, adaptively selecting the optimal neighborhood for spatial scanning by balancing spatial neighborhood information and local spatial structure. Subsequently, the SAES module dynamically enhances the spectral features of classified pixels using neighborhood spectral information and conducts spectral scanning. Finally, the spectral features of the target pixels are fed into a simple classifier to achieve high-precision HSI classification.

systems. These models transform input sequences $x(t) \in \mathbb{R}$ into output sequences $y(t) \in \mathbb{R}$ by employing a learnable latent state $h(t) \in \mathbb{R}^N$. The transformation follows these rules:

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B} \in \mathbb{R}^{N\times 1}$, and $\mathbf{C} \in \mathbb{D}^{1\times N}$ are the parameter matrices. To adapt the continuous system described in (1) for discrete data, the zero-order hold (ZOH) rule is employed to discretize the continuous parameters $(\mathbf{A}, \mathbf{B})$ into their discrete representations $(\overline{\mathbf{A}}, \overline{\mathbf{B}})$

$$\begin{aligned} \overline{\mathbf{A}} &= e^{\Delta\mathbf{A}} \\ \overline{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}(e^{\Delta\mathbf{A}} - \mathbf{I}) \cdot \Delta\mathbf{B} \end{aligned} \tag{2}$$

where Δ is a time scale parameter. After discretization, (1) transforms into

$$\begin{aligned} h_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t. \end{aligned} \tag{3}$$

The iterative calculation described by (3) can be efficiently performed in parallel using a global convolution operation

$$\begin{aligned} y &= x * \bar{\mathbf{K}} \\ \bar{\mathbf{K}} &= (\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \ldots, \mathbf{C}\bar{\mathbf{A}}^t\bar{\mathbf{B}}) \end{aligned} \tag{4}$$

where $*$ is convolution operation, and $\bar{\mathbf{K}}$ represents a convolution kernel with size $t + 1$.

B. Mamba

The traditional SSM achieves linear time complexity, but processing all inputs with the same parameters limits its effectiveness in capturing contextual sequences. To overcome this limitation, Mamba introduces a dynamic and selective mechanism. By treating SSM parameters as functions of the input, the model can adaptively change the parameters with the input, allowing for dynamic processing and selective forgetting of information based on the input context. Some work has begun to explore the application of Mamba in computer vision tasks. For instance, Vim utilizes bidirectional Mamba blocks to replace attention modules in the Vision Transformer (ViT) model. Moreover, VMamba develops 2-D Mamba modules for horizontal and vertical image scanning, forming a hierarchical visual model based on these modules. However, these models are primarily focused on the spatial characteristics of 2-D images, overlooking the spectral characteristics of high-dimensional HSIs.

III. Methodology

A. Overall Framework

In this article, we propose a novel HSI classification framework called HyperMamba, built upon the pure Mamba architecture. By adaptively scanning both spatial and spectral dimensions, our proposed model utilizes spatial neighborhood information to enhance the spectral features of classified pixels, achieving state-of-the-art HSI classification accuracy.

The overall framework is illustrated in Fig. 2. Specifically, given an HSI $\mathcal{H} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B represent the length, width, and number of spectral channels, respectively. PCA is then applied to the HSI data, resulting in $\mathcal{X} \in R^{H\times W\times C}$, with C indicating the reduced spectral dimension. Each pixel vector in $\mathcal{X}$ is treated as an individual sample, denoted as $x \in \mathbb{R}^{1\times 1\times C}$, and the entire dataset $\mathcal{X}$ is divided into a training subset $\mathcal{X}_{\text{train}}$ and a testing subset $\mathcal{X}_{\text{test}}$. Each sample x_i in the training subset is labeled with a one-hot label y. As suggested by Hu et al. [44] and Lv et al. [45], image patches centered around each classified pixel are generated for model input.

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To maximize the exploitation of spatial neighborhood information within HSIs during spatial scanning, we propose the SNAS module. This module dynamically adjusts the size of image patches during spatial scanning, guided by the spectral characteristics of classified pixels. This adaptive adjustment ensures an optimal balance between the amount of spatial neighborhood information and spatial structure within the neighborhood (see Section III-B for detailed information).

Next, we propose the SAES module for spectral scanning. This module dynamically enhances the spectral features of classified pixels by utilizing spectral information from neighboring pixels. Details are presented in Section III-C. Finally, the enhanced features are then fed into a simple linear classifier to generate the final model predictions. For training, we employ cross-entropy as the loss function, updating the model parameters by computing the loss between training samples and their labels, as shown in the following equation:

$$L_x = \frac{1}{B}\sum_{b=1}^{B} H(y_b,\, f(x_b)) \tag{5}$$

where $H(q, p)$ represents the cross-entropy between distributions q and p, $f(\cdot)$ denotes the proposed model, and B denotes the batch size.

B. Spatial Neighborhood Adaptive Scanning

To leverage spatial structure information for HSI classification, conventional methods utilize image patches centered on the classified pixels as input. However, Mamba models require converting 2-D image patches into 1-D sequences for spatial feature scanning. When processing excessively large image patches, the significant distances between adjacent pixels disrupt the spatial structure information within the patches. Conversely, employing smaller patches, while preserving local spatial dependence effectively, results in a reduction of neighborhood context information. To address this, we propose the SNAS module, dynamically adjusting patch size based on model input to effectively capture both local spatial relationships and neighborhood context during spatial scanning.

The details of the SNAS module are illustrated in Fig. 2. For the pixel requiring classification, we progressively crop its surrounding neighborhood using four different scales (h_1, w_1), (h_2, w_2), (h_3, w_3), and (h_4, w_4), yielding four differently sized image patches: M_1, M_2, M_3, and M_4. The spectral feature of the center classified pixel is then fed into a lightweight scale selection submodule containing three fully connected layers. Within this module, probabilities for selecting image patches at different scales, denoted as s, are computed. Subsequently, the patch with the highest probability, M_z, where $z = \arg\max(s)$, is chosen for subsequent feature extraction.

Due to the nondifferentiable nature of the scale selection process, the SNAS module cannot directly compute gradients for training. While evolutionary algorithms and zeroth-order optimization methods can optimize without gradient computation, they often converge slowly and struggle to achieve optimal performance. To achieve end-to-end model optimization, we employ the Gumbel–Softmax operation to provide a continuous approximation of the discrete decision process, as follows:

$$z'_k = \frac{\exp(\log(s_k + g_k)/\tau)}{\sum_{j=1}^{K}\exp(\log(s_j + g_j)/\tau)} \tag{6}$$

where K represents the number of selectable scales, $g_k = -\log(-\log(u_k))$, u_k is sampled from Uniform(0, 1), and τ is used to control the smoothness of z'_k. As τ tends to zero, z transforms into one-hot encoding, identical to OneHot(z). The selected image patch $M_{z'}$ is obtained as follows:

$$M_{z'} = \sum_{j=1}^{K} M_j \cdot z'_j. \tag{7}$$

In the practical implementation, for efficient batch-wise parallel processing, we employ region of interest (RoI) Pooling to downsample the selected $M_{z'}$ across different scales to a uniform scale. Next, we expand $M_{z'}$ into four 1-D sequences: top to bottom, bottom to top, left to right, and right to left. Then, each sequence is fed into the SSM block for spatial scanning independently. Finally, the obtained spatial features are combined and converted into a 2-D feature map $\mathcal{F}_{\text{spatial}}$.

C. Spectral Adaptive Enhancement Scanning

Current Mamba models neglect the inherent spectral characteristics of high-dimensional HSIs during feature scanning. Simply converting an HSI patch into a 1-D sequence can lead to interference between spectral and spatial information, disrupting both spectral information and spatial structure. In addition, excessively long sequences can result in significant computational costs. To address this issue, a novel SAES module is proposed. In this module, the spatial neighborhood information is utilized to enhance the spectral information of target classification pixels. Subsequently, the spectral features of classified pixels are scanned, achieving efficient and effective extraction of spectral information in HSIs.

During the spectral feature enhancement phase, based on the spatial features $\mathcal{F}_{\text{spatial}}$ extracted by the SNAS module, we compute the cosine similarity $s(f_{i,j} \cdot f_c)$ between each feature point $f_{i,j}$ in F_{spatial} and the central feature point f_c to obtain the similarity mask $\mathcal{P}$. Then, based on the similarity mask $\mathcal{P}$, we calculate the weighted average of the feature map $\mathcal{F}_{\text{spatial}}$ to obtain the enhanced spectral feature f_{spectral} for the classified pixel. The formulation for this process is as follows:

$$s(f_{i,j} \cdot f_c) = \frac{f_{i,j}^T \cdot f_c}{\|f_{i,j}\| \cdot \|f_c\|} \tag{8}$$

$$\mathcal{P}_{i,j} = \frac{s(f_{i,j} \cdot f_c)}{\max(s(f_{i,j})) - \min s(f_{i,j}))} \tag{9}$$

$$f_{\text{spectral}} = \text{MAP}(\mathcal{F}_{\text{spatial}} \odot \mathcal{P}) \tag{10}$$

where MAP represents the masked average pooling operation.

In the spectral feature scanning phase, the enhanced spectral feature f_{spectral} is sorted in both forward and backward directions to obtain f_{forward} and f_{backward}, respectively. These sorted features are individually fed into the SSM block for spectral scanning. Finally, the scanned spectral features are fused for classifying target pixels in HSIs.

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Overall, our method differs from existing Mamba approaches in two main aspects.

1) HyperMamba can adapt the scanning range dynamically based on the central pixel, unlike the fixed-range scanning used by previous methods [40], [41], [42]. In the field of computer vision, Mamba variants mainly focus on improving the scanning mechanism. For example, Vim uses forward and backward scanning to address scanning direction sensitivity, while VMamba expands scanning to four directions (horizontal and vertical axes). PlainMamba [46] proposed a continuous scan mechanism to reverse scan direction after each row or column to keep adjacent pixels, and EfficientVMamba [40] employs skip scanning to improve computational efficiency. Although these methods use different scanning techniques, they all transform entire 2-D images into 1-D sequences for processing. However, converting large image patches into 1-D sequences can disrupt spatial structure due to the significant distances between adjacent pixels. Smaller patches preserve local spatial dependencies but reduce neighborhood context. Different from these methods, our proposed SNAS module adaptively selects the optimal neighborhood for spatial scanning by balancing spatial neighborhood information with local spatial structure.
2) HyperMamba incorporates spatial–spectral information interaction during the scanning process. Current Mamba models neglect the inherent spectral characteristics of high-dimensional HSIs during feature scanning. Simply converting an HSI patch into a 1-D sequence can lead to interference between spectral and spatial information, disrupting both spectral information and spatial structure [36]. Unlike previous methods [40], [41], [42], HyperMamba addresses this issue by integrating spatial and spectral information interactions. The SNAS module adaptively selects spatial neighborhood ranges based on the spectral characteristics of classified pixels, while the SAES module dynamically enhances the spectral features of classified pixels and performs spectral scanning.

IV. Experiments

A. Data Description

To evaluate the effectiveness of our proposed method, we conduct extensive experiments on three widely used HSI datasets: the IP dataset, the Houston 2013 (HU) dataset, and the WHHH dataset [47]. The details of the datasets are provided below.

The IP dataset, collected in 1992 by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor, covers the IP agricultural site in Indiana, USA. The dataset comprises 145 × 145 pixels with a spatial resolution of 20 m, encompassing 220 spectral bands ranging from 400 to 2500 nm, with a spectral resolution of 10 m. After excluding 20 noisy and water absorption bands, the dataset retained 200 spectral bands specifically in bands 1–103, 109–149, and 164–219. It contains 16 land-cover types such as farmland, forests, urban areas, and water bodies. The false color image and the classification ground truth are shown in Fig. 3.

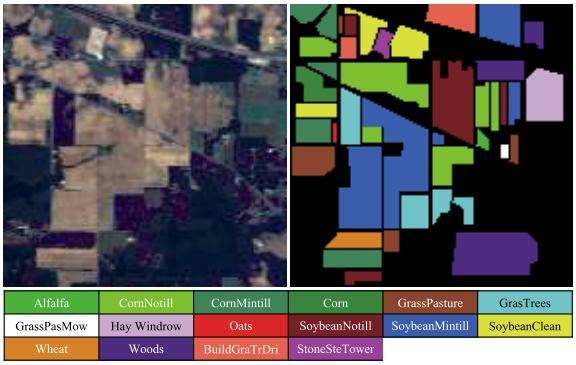


Fig. 3. Pseudo-color image and ground-truth map of the IP dataset.

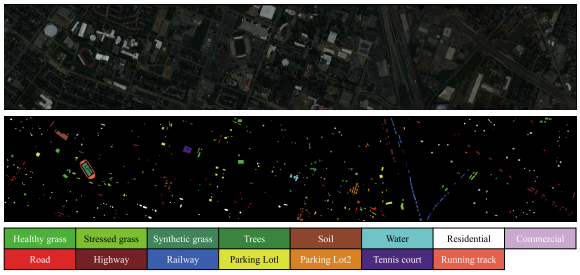


Fig. 4. Pseudo-color image and ground-truth map of the HU dataset.

The HU dataset, acquired in 2013 by the Compact Airborne Spectrographic Imager (CASI), encompasses both the University of Houston campus and its urban surroundings. It comprises 349 × 1905 pixels with a spatial resolution of 2.5 m per pixel and includes 144 spectral bands ranging from 380 to 1050 nm. The dataset categorizes land cover into 15 classes, such as Healthy grass and Railway. Fig. 4 provides a comprehensive visualization of the dataset, displaying both the false-color image and the ground-truth map.

The WHHH dataset was captured on November 20, 2017, in Honghu City, Hubei Province, China. It was obtained using a Headwall Nano-Hyperspec imaging sensor mounted on a DJI Matrice 600 Pro unmanned aerial vehicle (UAV) platform flying at an altitude of 100 m. The imagery is 940 × 475 pixels in size, with a spatial resolution of approximately 0.043 m, covering 270 bands from 400 to 1000 nm. The dataset includes 17 crop types, with variations in cultivars like Chinese cabbage/cabbage and Brassica chinensis/small Brassica chinensis, totaling 22 different land-cover classes. Refer to Fig. 5 for the false-color image and classification ground truth.

B. Experimental Settings

1) Dataset Settings: Following prior protocols [48], we construct a training subset for each dataset by selecting ten samples from each land-cover category. The remaining samples are allocated to the testing subset. Table I summarizes the sizes of both the training and testing datasets, detailing the distribution of samples across different land-cover categories.

2) Training Details: To ensure a fair comparison, we replicate the optimizer and learning rate settings from spectral query spatial transformer (SQSFormer), utilizing the Adam optimizer with a learning rate of $1e^{-3}$. In addition, we set

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TABLE I
CLASS-WISE DISTRIBUTION OF TRAINING AND TESTING SAMPLES ON THE IP, HU, AND WHHH DATASETS

No.	Indian Pines			Houston 2013			WHU-Hi-HongHu		
	Class	Train	Test	Class	Train	Test	Class	Train	Test
1	Alfalfa	10	36	Healthy grass	10	1241	Red roof	10	14031
2	CornMotill	10	1418	Stressed grass	10	1244	Road	10	3502
3	CornMintill	10	820	Synthetic grass	10	687	Bare soil	10	21811
4	Corn	10	227	Trees	10	1234	Cotton	10	163275
5	GrassPasture	10	473	Soil	10	1232	Cotton firewood	10	6208
6	GrassTrees	10	720	Water	10	315	Rape	10	44547
7	GrassPastureMowed	10	18	Residential	10	1258	Chinese cabbage	10	24093
8	Hay Windrowed	10	468	Commercial	10	1234	Pakchoi	10	4044
9	Oats	10	10	Road	10	1242	Cabbage	10	10809
10	SoybeanNotill	10	962	Highway	10	1217	Tuber mustard	10	12384
11	SoybeanMintill	10	2445	Railway	10	1225	Brassica parachinensis	10	11005
12	SoybeanClean	10	583	Parking Lot1	10	1223	Brassica chinensis	10	8944
13	Wheat	10	195	Parking Lot2	10	459	Small Brassica chinensis	10	22497
14	Woods	10	1255	Tennis court	10	418	Lactuca sativa	10	7346
15	BuildingsGrassTreesDrives	10	376	Running track	10	650	Celtuce	10	992
16	StoneSteelTowers	10	83				Film covered lettuce	10	7252
17							Romaine lettuce	10	3000
18							Carrot	10	3207
19							White radish	10	8702
20							Garlic sprout	10	3476
21							Broad bean	10	1318
22							Tree	10	4030
	Total	160	10089	Total	150	14879	Total	220	386473

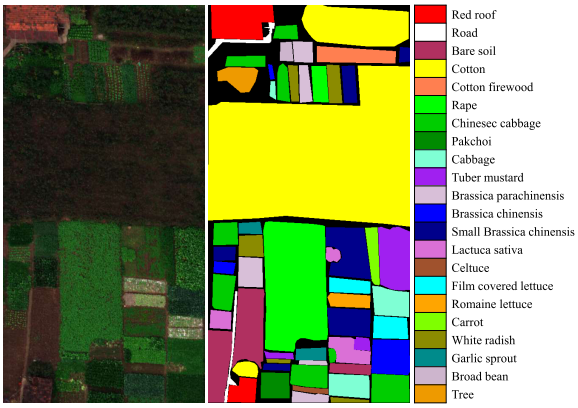


Fig. 5. Pseudo-color image and ground-truth map of the WHHH dataset.

a consistent batch size of 64 for the IP, HU, and WHHH datasets. The PCA channel numbers for these datasets are set to 200, 20, and 30, respectively. For the neighborhood scales (h_1, w_1), (h_2, w_2), (h_3, w_3), and (h_4, w_4), they are set as $(13, 13)$, $(11, 11)$, $(9, 9)$, and $(7, 7)$, respectively, on the IP and HU datasets, and $(21, 21)$, $(19, 19)$, $(17, 17)$, and $(15, 15)$, respectively, on the WHHH dataset. τ is set to 1. All experiments are conducted on a server equipped with three NVIDIA A100 GPUs, using the PyTorch framework.

3) Models and Evaluation Metrics for Comparison: We conducted comparative experiments with 11 state-of-the-art models to comprehensively assess the effectiveness of our proposed method.

1) *Spectral–Spatial Residual Network (SSRN) [49]:* The SSRN effectively extracts and fuses spectral–spatial information, making it one of the most advanced CNN-based methods for HSI classification.
2) *SpectralFormer [32]:* SpectralFormer captures local spectral representations from multiple neighboring bands in HS images, rather than just a single band. It is a classic Transformer-based method for HSI classification.
3) *Spatial–Spectral Involution Multilayer Perceptron Network (SSIN) [50]:* SSIN is a multilayer perceptron (MLP)-based model for hyperspectral classification. It features two extraction paths: the image path employs InvoMLP blocks for spatial–spectral feature extraction, while the coordinate path enhances data diversity. This dual approach improves feature robustness and classification performance.
4) *Hyperspectral Image Transformer (HiT) [26]:* By integrating convolutional operations into a Transformer architecture, HiT effectively captures subtle spectral differences and conveys local spatial context information, leading to promising performance across various HSI classification datasets.
5) *SSGRN [25]:* SSGRN is a GCN-based method for hyperspectral classification that extracts spatial and spectral contexts through two subnetworks: spatial graph reasoning subnetwork (SAGRN) and spectral graph reasoning subnetwork (SEGRN). SSGRN is finally produced by combining SAGRN and SEGRN to improve the classification.
6) *Dual-View Spectral and Global Spatial Feature Fusion Network (DSGSF) [51]:* DSGSF is an LSTM-based hyperspectral classification network. It consists of a spatial subnetwork, which utilizes an encoder–decoder structure with channel and spatial attention to capture global spatial features, and a spectral subnetwork that employs a dual-view spectral feature aggregation model with view attention to enhance spectral diversity.
7) *Spatial–Spectral 1D Swin (SS1DSwin) [52]:* SS1DSwin models local and hierarchical spatial–spectral relationships to construct discriminative representations, enhancing HSI classification performance.

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TABLE II

EXPERIMENTAL RESULTS ON THE IP DATASET: COMPARISON WITH STATE-OF-THE-ART METHODS (MEAN AND STANDARD DEVIATION FROM FIVE RUNS), HIGHLIGHTING BEST IN BLACK AND SECOND BEST IN BLUE

Class	SSRN (2018)	SpectralFormer (2022)	SSIN (2022)	HiT (2022)	SSGRN (2023)	DSGSF (2023)	SS1DSwin (2023)	SSFTT (2022)	CASST (2022)	LSGA (2023)	SQSFormer (2024)	Ours
1	**100.00±0.00**	87.22±11.21	89.89±1.36	**99.44±1.24**	97.78±3.24	58.20±37.03	88.89±5.20	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**99.44±1.24**	**100.00±0.00**
2	70.89±12.57	24.30±4.44	78.22±1.41	40.42±11.73	62.07±6.85	50.37±11.98	38.60±4.32	75.98±2.67	**80.37±2.58**	49.37±3.59	78.87±2.06	**82.02±4.51**
3	77.95±3.27	52.07±5.52	77.46±7.73	31.17±6.67	64.88±9.78	38.26±22.68	62.44±12.55	**84.63±1.58**	80.15±3.19	84.17±1.81	82.56±1.26	**89.20±5.29**
4	97.71±1.75	55.77±13.32	97.27±1.23	63.35±24.62	95.51±4.15	37.64±11.00	61.06±11.50	93.30±3.83	89.07±2.30	**99.91±0.20**	**99.21±0.65**	96.74±1.52
5	93.78±3.05	50.15±15.23	83.68±4.96	38.09±4.69	82.83±5.36	70.22±17.67	79.20±10.62	**94.71±1.11**	92.56±1.13	84.14±1.90	**96.62±1.46**	86.30±6.86
6	94.08±1.66	54.92±3.79	95.39±1.91	73.22±9.47	93.75±3.48	75.26±13.32	63.33±29.82	93.36±0.58	**96.50+1.96**	95.83±1.47	94.31±0.86	**98.25±0.93**
7	**100.00±0.00**	87.78±15.42	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	29.86±14.91	**88.89±8.78**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	**99.96±0.10**	82.39±8.86	**100.00±0.00**	98.58+1.31	99.19±1.03	94.25±2.56	92.69±2.03	**100.00±0.00**	**100.00±0.00**	99.91±0.19	**100.00±0.00**	**100.00±0.00**
9	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**98.00±4.47**	**100.00±0.00**	22.46±13.90	96.00±8.94	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	86.51±2.42	52.93±9.01	82.68±2.58	60.09±16.08	79.42±4.36	50.02±19.60	56.11±4.21	79.83±3.65	75.24±1.48	79.50±2.66	**87.61±2.10**	**89.69±4.26**
11	68.76±5.47	32.25±10.66	**78.31±5.43**	56.92±9.22	69.87±3.05	60.96±13.77	44.79±7.99	67.46±4.00	65.43±1.36	69.22±3.73	75.43±3.27	**82.93±6.01**
12	87.48±2.20	26.52±6.36	88.58±5.29	32.71±10.14	71.90±5.61	40.14±17.72	51.70±7.91	81.37±6.01	81.17±5.04	81.06±5.54	**93.17±0.89**	**94.27±2.88**
13	**99.28±0.78**	74.36±21.55	98.77±1.15	81.54±5.54	98.77±1.06	80.81±10.55	97.13±0.58	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.08±1.14
14	98.15±0.99	62.71±22.16	**98.25±2.36**	84.01±16.40	95.03±2.43	79.40±9.62	83.38±5.89	97.91±0.51	95.38±2.29	89.96±2.24	**98.73±0.16**	97.35±2.71
15	96.06±3.01	38.56±10.61	**98.72±1.13**	44.30±8.24	98.09±2.96	51.63±14.90	78.14±12.30	98.35±0.55	96.86±1.57	85.21±4.42	**99.15±0.61**	98.09±2.58
16	98.80±2.09	97.59±2.95	**100.00±0.00**	93.49±4.48	97.83±3.76	69.20±11.53	98.55±1.98	**100.00±0.00**	**99.28±1.62**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
OA (%)	83.37±1.88	45.44±3.35	86.09±1.28	58.79±4.74	78.80±2.77	58.44±11.96	60.14±5.04	83.30±1.35	82.26±0.59	78.24±0.92	**87.31±0.83**	**89.88±1.22**
AA (%)	91.84±0.78	61.22±1.85	92.26±0.53	68.46±3.02	87.93±1.90	56.79±9.41	73.81±3.36	91.68±0.63	90.75±0.57	88.64±0.60	**94.07±0.29**	**94.72±0.80**
$\kappa \times 100$	81.31±2.07	39.90±3.41	84.29±1.41	53.24±5.16	76.16±3.11	52.43±13.93	55.64±5.48	81.20±1.49	80.04±0.65	75.41±1.00	**85.67±0.92**	**88.54±1.36**

TABLE III

EXPERIMENTAL RESULTS ON THE HU DATASET: COMPARISON WITH STATE-OF-THE-ART METHODS (MEAN AND STANDARD DEVIATION FROM FIVE RUNS), HIGHLIGHTING BEST IN BLACK AND SECOND BEST IN BLUE

Class	SSRN (2018)	SpectralFormer (2022)	SSIN (2022)	HiT (2022)	SSGRN (2022)	DSGSF (2022)	SS1DSwin (2023)	SSFTT (2022)	CASST (2022)	LSGA (2023)	SQSFormer (2024)	Ours
1	82.98±4.75	87.22±2.01	86.06±7.06	89.30±3.08	57.03±6.30	91.52±4.29	**96.15±1.03**	90.64±5.30	87.11±5.05	90.56±4.59	89.27±7.69	**92.76±4.22**
2	85.95±4.97	94.74±2.88	90.63±4.47	94.53±1.06	39.50±6.08	92.29±2.91	92.04±3.88	94.10±1.75	92.09±2.01	96.83±1.59	**95.88±2.79**	**97.22±0.85**
3	97.76±2.80	98.40±1.31	**99.45±0.52**	96.80±2.55	92.08±3.90	93.69±5.64	97.52±2.20	96.80±2.91	97.26±3.10	98.92±1.66	99.36±0.65	**99.91±0.12**
4	85.66±3.82	76.30±8.92	83.86±1.97	89.66±4.22	25.36±5.27	**94.59±2.62**	86.22±5.87	88.65±8.59	87.34±5.59	**92.66±3.78**	90.65±3.44	92.54±4.43
5	99.55±0.83	98.86±0.93	**99.98±0.03**	99.12±1.75	55.67±4.47	91.71±5.15	93.56±3.73	99.64±0.42	99.92±0.16	98.73±2.53	99.97±0.04	**100.00±0.00**
6	91.62±4.62	83.81±13.65	94.10±4.80	93.40±4.77	83.30±6.92	94.90±1.72	85.20±3.53	92.32±5.68	89.14±4.08	**96.38±4.65**	90.80±4.19	**96.19±4.54**
7	83.43±7.08	82.72±2.49	83.13±6.87	76.13±6.70	37.28±5.44	79.58±5.06	77.07±4.08	77.20±3.96	81.14±4.86	**88.09±3.79**	84.44±5.84	**89.46±2.53**
8	50.41±7.88	63.27±6.21	55.54±8.19	51.53±7.27	23.50±3.96	**74.26±7.79**	52.22±1.85	51.43±5.68	**65.09±11.82**	60.57±9.01	60.45±9.24	63.94±4.63
9	76.65±4.16	63.96±6.59	73.33±12.25	65.62±3.54	29.10±2.16	69.27±6.04	58.63±12.43	69.07±10.20	67.67±7.44	74.91±3.74	**77.52±8.21**	**77.02±3.32**
10	**91.69±10.53**	72.64±6.30	**97.47±3.26**	86.72±6.87	35.45±3.77	70.45±7.35	73.87±6.28	90.55±10.20	84.29±10.17	91.52±12.21	87.64±10.62	90.98±11.27
11	90.78±7.32	65.01±11.25	**95.87±3.97**	77.67±14.18	34.11±4.38	75.33±6.45	72.02±4.17	84.54±9.65	**94.40±2.53**	90.22±4.99	89.00±6.14	91.00±4.16
12	64.63±15.00	**80.67±8.85**	71.41±7.27	78.79±14.09	39.30±6.01	58.24±8.59	56.37±14.70	78.10±11.85	**85.07±9.94**	70.76±11.07	79.69±10.16	69.47±14.81
13	91.37±4.71	70.28±6.04	93.55±3.03	86.84±3.34	67.02±6.55	41.78±10.78	61.05±7.73	**94.56±3.75**	90.02±2.97	**96.47±3.33**	90.33±3.79	91.55±2.89
14	**100.00±0.00**	98.23±0.96	**100.00±0.00**	99.81±0.28	**99.95±0.10**	79.43±12.31	98.90±0.54	**100.00±0.00**	99.47±0.83	**100.00±0.00**	99.86±0.19	99.47±0.75
15	99.38±0.95	95.08±2.49	**100.00±0.00**	**100.00±0.00**	69.26±10.14	94.60±1.97	98.89±0.77	**99.97±0.06**	99.38±0.78	**100.00±0.00**	**100.00±0.00**	**99.97±0.06**
OA(%)	83.79±0.54	80.60±1.62	86.13±0.55	83.49±1.44	45.16±1.96	79.85±1.87	78.25±1.09	84.89±1.46	86.34±2.07	**87.72±1.09**	87.40±1.61	**88.40±1.15**
AA(%)	86.12±0.68	82.07±1.29	88.29±0.64	85.73±1.35	52.53±1.76	80.11±2.06	79.98±0.94	87.17±1.21	87.96±1.62	**89.78±0.85**	88.99±1.36	**90.10±0.95**
$\kappa \times 100$	82.51±0.59	79.04±1.75	85.03±0.59	82.18±1.55	41.06±2.05	78.23±2.01	76.53±1.18	83.68±1.57	85.24±2.24	**86.74±1.18**	86.39±1.74	**87.47±1.24**

8) *Spectral–Spatial Feature Tokenization Transformer (SSFTT) [53]:* SSFTT is a hybrid CNN–Transformer model for HSI classification. It extracts shallow features with the CNN and high-level semantic features with Transformer encoders, achieving competitive performance across multiple HSI datasets.
9) *CASST [33]:* CASST is a dual-branch Transformer, with one branch focusing on spatial features and the other on spectral features. By integrating spatial and spectral features through cross-attention, it performs well on various HSI datasets.
10) *Light Self-Gaussian Attention (LSGA) [54]:* A recent Transformer-based hyperspectral classification algorithm, LSGA, integrates an LSGA module into the ViT model, enhancing its ability to extract global deep semantic features and demonstrating strong performance across various HSI datasets.
11) *SQSFormer [48]:* SQSFormer is one of the most recent Transformer-based algorithms. It adaptively extracts spatial information from neighboring pixels based on the central pixel, achieving state-of-the-art performance across multiple HSI datasets.

The performance of the above methods is compared with the proposed method on four metrics: overall accuracy (OA), average accuracy (AA), Kappa coefficient ($\kappa \times 100$), and per-class classification accuracy.

C. Results Comparison and Analysis

Tables II–IV report the mean and standard deviation of OA, AA, $\kappa \times 100$, and per-class classification accuracy across five replicate experiments conducted on the IP, HU, and WHHH datasets for the aforementioned models.

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TABLE IV
EXPERIMENTAL RESULTS ON THE WHHH DATASET: COMPARISON WITH STATE-OF-THE-ART METHODS (MEAN AND STANDARD DEVIATION FROM FIVE RUNS), HIGHLIGHTING BEST IN BLACK AND SECOND BEST IN BLUE

Class	SSRN (2018)	SpectralFormer (2022)	SSIN (2022)	HiT (2022)	SSGRN (2023)	DSGSF (2023)	SS1DSwin (2023)	SSFTT (2022)	CASST (2022)	LSGA (2023)	SQSFormer (2024)	Ours
1	**94.71±1.68**	77.43±0.89	90.96±5.18	**93.93±1.16**	72.49±3.66	93.79±3.62	88.89±3.13	91.93±1.04	93.03±0.49	93.88±1.69	88.99±2.36	91.78±4.37
2	79.26±7.25	52.20±6.45	76.04±8.83	73.79±6.87	50.91±6.22	48.26±7.12	82.07±1.31	81.52±4.43	78.66±2.88	**87.11±2.51**	**95.55±1.28**	79.81±12.63
3	83.96±1.92	78.44±3.08	**87.61±4.01**	78.99±3.86	46.06±11.23	83.32±8.98	72.94±1.71	80.72±0.68	82.57±0.45	82.16±2.03	81.53±3.20	**87.34±4.37**
4	97.71±1.46	69.03±7.81	91.22±3.60	92.16±0.75	64.94±10.17	98.51±0.56	84.48±6.84	90.73±1.06	93.12±0.78	90.59±2.21	**98.00±1.16**	**98.68±0.49**
5	90.68±3.82	67.44±0.62	**96.91±2.80**	86.60±2.55	95.10±2.17	32.64±16.25	70.61±10.30	92.68±2.32	93.12±1.39	92.28±2.06	94.57±1.67	**95.70±1.58**
6	89.36±3.51	65.68±1.30	**95.08±1.37**	83.80±2.79	33.96±9.49	90.45±1.42	79.27±4.31	91.84±0.74	91.96±0.76	91.84±0.72	**97.25±1.13**	94.87±0.83
7	63.90±15.99	35.27±1.68	73.83±8.50	50.72±3.38	29.64±4.11	**76.52±10.46**	52.45±7.26	66.44±2.71	68.16±4.33	68.28±2.75	73.62±2.52	**77.13±9.94**
8	77.50±7.51	23.29±2.17	**96.59±2.84**	33.90±15.23	89.38±7.33	20.84±6.52	35.30±2.90	65.44±3.07	69.02±2.02	84.07±1.56	90.45±1.44	**90.61±2.46**
9	93.59±1.08	88.74±1.70	**96.05±2.34**	82.86±8.08	74.22±9.02	91.18±7.12	88.12±7.64	93.85±0.87	94.79±0.08	94.13±0.29	94.07±0.75	**96.79±2.33**
10	80.30±6.94	37.48±10.71	82.31±2.58	33.34±9.86	45.29±5.55	48.12±11.38	53.81±10.71	70.90±7.99	78.94±1.00	75.44±3.64	**84.95±3.35**	**88.28±1.92**
11	51.38±18.29	28.04±2.21	**74.68±3.82**	23.10±5.77	44.33±5.51	45.90±10.13	46.22±1.29	61.42±3.28	66.52±1.32	70.13±5.55	67.93±3.34	**84.90±2.87**
12	67.82±20.81	56.88±5.24	**77.04±4.04**	60.77±5.87	58.58±7.12	39.20±12.63	54.16±3.00	58.37±5.38	70.54±1.82	64.43±3.52	**88.23±1.94**	75.40±3.73
13	55.45±28.57	34.53±4.64	66.47±3.65	47.01±5.83	33.52±7.00	67.01±11.12	48.43±4.08	48.20±7.27	43.28±3.87	61.22±4.73	**69.80±6.52**	**69.95±5.46**
14	92.90±5.64	57.56±2.18	84.21±5.76	60.74±5.73	55.48±7.34	51.73±12.65	70.60±9.26	92.56±2.38	**96.20±0.63**	68.95±1.08	**96.57±1.40**	92.24±1.57
15	**99.42±0.84**	74.54±3.70	96.15±3.27	92.24±2.56	93.02±5.62	32.26±18.73	86.36±5.32	98.13±1.11	**98.19±0.61**	96.83±1.07	97.32±0.20	95.87±1.75
16	87.37±27.02	96.70±2.16	57.76±9.71	89.19±4.87	27.08±9.00	83.97±7.96	68.34±5.34	90.44±1.62	84.22±1.64	**97.23±1.89**	89.20±4.81	**91.31±4.43**
17	**99.56±0.39**	71.41±3.52	93.71±7.23	74.27±10.87	98.15±1.31	72.95±15.75	74.02±3.22	96.87±0.85	96.54±0.60	95.03±0.64	**99.83±0.23**	93.39±7.44
18	**99.42±0.56**	33.93±18.76	96.58±1.56	63.04±10.48	66.64±13.30	60.73±13.94	70.23±2.80	**99.40±0.34**	99.19±0.30	98.30±0.50	96.99±1.13	98.26±0.75
19	86.73±10.97	67.34±5.83	85.03±4.91	57.44±7.63	50.70±3.93	68.76±6.56	78.81±2.47	89.43±1.84	**90.67±2.54**	85.73±4.47	82.51±2.01	**92.76±2.68**
20	**98.04±0.97**	77.72±4.09	95.68±1.71	74.39±1.74	89.28±5.43	46.67±16.07	80.84±11.67	83.14±5.55	90.34±1.32	**96.67±0.53**	**98.04±1.11**	96.41±1.50
21	98.82±1.14	49.88±4.17	**99.70±0.42**	63.57±10.32	88.82±6.72	18.03±9.20	87.08±2.01	91.53±2.88	93.12±3.38	94.83±1.61	98.98±0.97	**99.77±0.28**
22	90.67±6.39	53.72±3.84	**99.69±0.20**	86.23±4.55	98.16±2.42	38.15±17.11	78.00±4.42	86.53±5.03	88.32±0.84	95.23±1.80	**99.96±0.04**	96.21±4.50
OA(%)	87.55±4.48	61.93±4.44	87.91±1.06	76.31±1.34	56.90±5.52	74.06±7.05	74.88±4.45	83.93±0.36	85.71±0.43	85.47±1.64	**91.22±0.50**	**92.23±0.50**
AA(%)	85.39±5.90	57.20±1.29	88.40±0.79	65.45±1.75	66.48±2.61	60.08±6.33	70.50±2.80	82.82±0.68	84.57±0.39	85.65±1.07	**90.20±0.35**	**90.34±0.55**
$\kappa \times 100$	84.35±5.68	55.18±4.43	85.02±1.21	70.65±1.66	49.80±5.53	68.75±7.77	69.45±5.01	80.10±0.42	82.24±0.52	82.02±1.95	**88.95±0.61**	**90.20±0.63**

1) Comparison on the IP Dataset: Table II presents a comparison between HyperMamba and representative methods for HSI classification on the IP dataset. HyperMamba is significantly superior to other state-of-the-art methods, achieving OA, AA, and $\kappa \times 100$ of 89.88%, 94.72%, and 88.54%, respectively. Compared to the second-best method, SQSFormer, HyperMamba shows improvements of 2.57%, 0.65%, and 2.87% in OA, AA, and $\kappa \times 100$, respectively. It is worth noting that our method achieves over 80% accuracy for each class. Particularly notable is the substantial 7.50% accuracy improvement for the previously challenging "SoybeanMintill" class (Class 11) compared to the second-best method, SQSFormer, increasing from 75.43% to 82.93%.

2) Comparison on the HU Dataset: Table III reports the promising performance of the proposed HyperMamba on the HU dataset. HyperMamba achieves OA, AA, and $\kappa \times 100$ of 88.40%, 90.10%, and 87.47%, respectively, consistently outperforming the previous state-of-the-art Transformer-based method, LSGA. Specifically, our method improves OA performance by 0.68% compared to LSGA, with AA increasing by 0.32% and $\kappa \times 100$ by 0.73%. In addition, HyperMamba demonstrates excellent per-class accuracy, exceeding 90% accuracy in 11 out of 15 categories, which highlights its effectiveness in handling diverse classes within the HU dataset.

3) Comparison on the WHHH Dataset: As reported in Table IV, on the challenging WHHH dataset, our proposed HyperMamba still achieves promising performance. Specifically, HyperMamba achieves OA, AA, and $\kappa \times 100$ of 92.32%, 90.34%, and 90.20%, respectively, surpassing the second-best method, SQSFormer, by 1.01%, 0.14%, and 1.25%, respectively. Among the 22 categories, HyperMamba achieves the highest accuracy in 11 of them. Moreover, for the 11th class, "Brassica chinensis," HyperMamba exhibits a notable accuracy improvement of 14.77%, surpassing the second-best performing model, LSGA. Specifically, accuracy increases from 70.13% to 84.90%.

D. Parameter and Computational Complexity Analysis

Table V presents a comprehensive comparison of model parameters and computational complexity between HyperMamba and alternative methods, with computational complexity assessed in terms of FLOPs. For example, in the IP dataset, HyperMamba exhibits the lowest parameter count and computational complexity, requiring only 199.61k parameters and 10.59M FLOPs, respectively. Competitive results are also observed in the HU and WHHH datasets, where HyperMamba demands merely 159.19k parameters and 9.72M FLOPs for the former, and 162.11k parameters and 9.76M FLOPs for the latter. To provide a more intuitive illustration of the benefits of our approach, we present a comprehensive comparison of model parameters and OA in Fig. 6. Our method achieves the highest accuracy while requiring remarkably low parameter quantities and computational complexity. These findings underscore the practicality of HyperMamba for scenarios with limited resources or real-time processing requirements. In addition, we present the inference times of the model on the test sets for the IP, HU, and WHHH datasets in Table VI. Our method not only achieves the highest classification accuracy but also shows competitive performance in terms of inference speed, with an inference time 89.64 s shorter than the classical HiT method on the WHHH dataset. This further substantiates the practicality of our approach.

E. Ablation Study

1) Effects of Different Modules: To further validate the efficacy of the proposed approach, ablative experiments are

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TABLE V
COMPARISON OF MODEL PARAMETERS AND COMPUTATIONAL COMPLEXITY

Methods	Indian Pines		Houston2013		WHU-Hi-HongHu	
	FLOPs (M)	Parameters (k)	FLOPs (M)	Parameters (k)	FLOPs (M)	Parameters (k)
SSRN	358.81	540.81	29.18	74.40	123.94	100.32
SpectralFormer	50.48	378.55	2.83	126.73	4.39	130.98
SSIN	53.74	1099.39	52.34	1082.01	52.42	1083.65
HiT	873.09	59287.24	612.08	52751.28	1617.57	52870.56
SSGRN	14345.90	2948.37	73482.95	16103.72	308041.41	193044.88
DSGSF	753.57	442.74	19047.16	429.36	13011.09	446.51
SS1DSwin	111.54	441.81	23.20	391.39	111.54	442.20
SSFTT	79.94	936.58	7.70	107.53	41.10	153.61
LSGA	231.55	1443.02	96.60	470.90	386.80	591.02
CASST	2700.83	13804.62	1285.99	12889.17	3466.51	13079.64
SQSFormer	440.45	1346.63	391.78	1059.80	1950.66	1110.61
Ours	10.59	199.61	9.72	159.19	9.76	162.11

TABLE VI
COMPARISON OF TEST TIME BETWEEN OUR METHOD AND THE COMPARISON METHODS ON THE IP, HU, AND WHHH DATASETS, WITH THE BEST RESULTS HIGHLIGHTED IN BOLD

Test Time	SSRN	SpectralFormer	SSIN	HiT	SSGRN	DSGSF	SS1DSwin	SSFTT	CASST	LSGA	SQSFormer	Our
IP	0.38	1.91	3.33	2.99	**0.15**	1.20	2.62	0.40	11.40	2.26	0.66	2.88
HU	0.39	2.38	1.83	4.36	1.62	1.97	3.71	**0.34**	2.28	2.30	0.69	0.82
WHHH	11.46	22.11	47.40	113.09	**0.64**	1.86	72.84	9.63	82.30	12.55	19.31	23.45

TABLE VII
ABLATION EXPERIMENTS ON THE TWO CORE MODULES OF HYPERMAMBA (SNAS AND SAES) ACROSS THE IP, HU, AND WHHH DATASETS, WITH BEST PERFORMANCE HIGHLIGHTED IN BOLD

Baseline	SNAS	SAES	Indian Pines			Houston2013			WHU-Hi-HongHu		
			OA	AA	$\kappa \times 100$	OA	AA	$\kappa \times 100$	OA	AA	$\kappa \times 100$
✓	-	-	84.64±1.37	92.19±0.21	82.71±1.51	87.55±0.54	89.43±0.53	86.56±0.58	91.20±0.29	89.57±0.52	89.04±0.35
✓	✓	-	87.36±2.03	93.05±0.67	85.74±2.24	87.93±1.05	89.73±0.87	86.98±1.13	91.80±0.46	90.00±0.47	89.62±0.66
✓	-	✓	85.40±0.82	92.38±0.24	83.51±0.90	88.09±1.49	89.90±1.14	87.13±1.61	91.48±0.17	89.73±0.47	89.39±0.31
✓	✓	✓	**89.88±1.22**	**94.72±0.80**	**88.54±1.36**	**88.40±1.15**	**90.10±0.95**	**87.47±1.24**	**92.23±0.50**	**90.34±0.55**	**90.20±0.63**

TABLE VIII
IMPACT OF NUMBER OF SELECTABLE SCALES IN THE SCALE SELECTION SUBMODULE, WITH THE BEST PERFORMANCE HIGHLIGHTED IN BOLD

Number of scales	Indian Pines			Houston2013			WHU-Hi-HongHu		
	OA	AA	$\kappa \times 100$	OA	AA	$\kappa \times 100$	OA	AA	$\kappa \times 100$
1	85.40±0.82	92.38±0.24	83.51±0.90	88.09±1.49	89.90±1.14	87.13±1.61	91.48±0.17	89.73±0.47	89.39±0.31
2	88.89±1.41	94.21±0.48	87.44±1.55	88.18±1.46	89.75±1.19	87.24±1.57	91.95±0.27	90.20±0.40	89.86±0.35
3	89.05±1.34	94.13±0.31	87.61±1.48	88.27±1.57	89.87±1.26	87.33±1.70	92.07±0.29	90.33±0.27	90.01±0.35
4	**89.88±1.22**	**94.72±0.80**	**88.54±1.36**	**88.40±1.15**	**90.10±0.95**	**87.47±1.24**	**92.23±0.50**	**90.34±0.55**	**90.20±0.63**

conducted on three datasets: IP, HU, and WHHH. The aim is to analyze the influence of two core modules, SNAS and SAES, on model performance, as depicted in Table VII. Upon removing both the SNAS and SAES modules, the baseline model achieves OA of 84.64%, 87.55%, and 91.20% on the IP, HU, and WHHH datasets, respectively. When incorporating the SNAS module into the baseline model, we observe accuracy improvements of 2.72%, 0.38%, and 0.60% on the IP, HU, and WHHH datasets, respectively. This indicates that the SNAS module effectively maintains spatial structure while incorporating neighborhood information through adaptive selection of spatial neighborhood size.

Similarly, upon integrating the SAES module into the baseline model, we observe OA improvements of 0.76%, 0.54%, and 0.28% on the IP, HU, and WHHH datasets, respectively, demonstrating the effectiveness of the SAES mechanism. Notably, the integration of both modules yields greater performance gains, with improvements of 5.24%, 0.85%, and 1.03% on the IP, HU, and WHHH datasets, respectively, compared to the baseline model. This confirms the complementarity of the two modules.

2) Effects of Number of Selectable Scales in the Scale Selection Submodule: We further analyzed the effectiveness of some key submodules within the SNAS and SAES modules. In Table VIII, we evaluated the effect of varying the number of selectable scales from 1 to 4 in the SNAS module on model performance across the IP, HU, and WHHH datasets. As the number of image blocks increased, model performance consistently improved. This demonstrates that adaptively selecting the spatial neighborhood scanning range helps balance spatial neighborhood information and structural preservation in our method.

3) Effects of Spectral Scanning and Spectral Enhancement in the SEAS Module: In Table IX, we evaluated the effects

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TABLE IX

IMPACT OF SPECTRAL SCANNING AND SPECTRAL ENHANCEMENT IN THE SEAS MODULE, WITH THE BEST PERFORMANCE HIGHLIGHTED IN BOLD

SEAS module	Indian Pines			Houston2013			WHU-Hi-HongHu		
	OA	AA	$\kappa \times 100$	OA	AA	$\kappa \times 100$	OA	AA	$\kappa \times 100$
Base Model	87.36±2.03	93.05±0.67	85.74±2.24	87.93±1.05	89.73±0.87	86.98±1.13	91.80±0.46	90.00±0.47	89.62±0.66
+ Spectral Scanning	87.95±1.64	93.27±0.49	86.38±1.81	88.14±1.11	89.82±0.81	87.19±1.19	91.97±0.16	89.85±0.30	89.87±0.22
+ Spectral Enhancement	88.59±1.78	93.91±0.53	87.09±1.97	88.29±1.17	89.96±0.89	87.35±1.26	92.07±0.68	89.97±0.49	90.00±0.86
Our	**89.88±1.22**	**94.72±0.80**	**88.54±1.36**	**88.40±1.15**	**90.10±0.95**	**87.47±1.24**	**92.23±0.50**	**90.34±0.55**	**90.20±0.63**

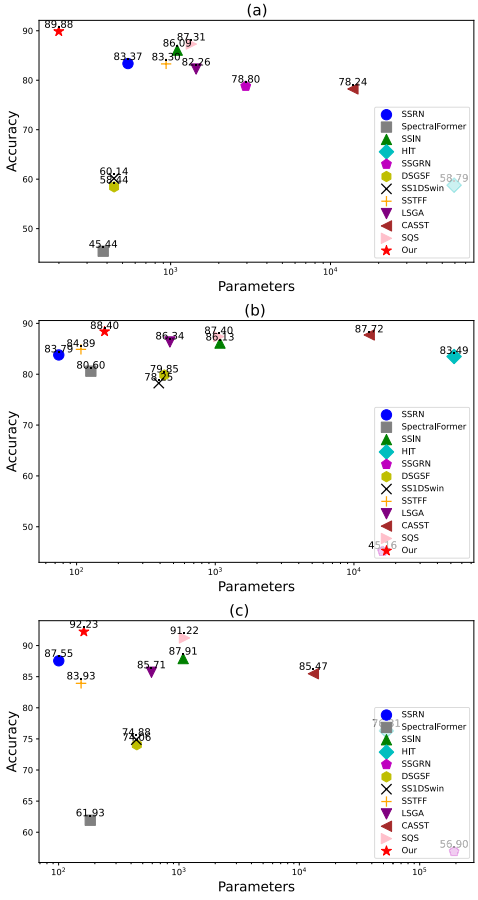


Fig. 6. OA and parameters of different methods on (a) IP, (b) HU, and (c) WHHH datasets, respectively.

of spectral scanning and spectral enhancement within the SEAS module on model performance. Both operations led to performance improvements. Spectral scanning provided gains of 0.59%, 0.21%, and 0.17% across the three datasets compared to the baseline, while spectral enhancement resulted in improvements of 1.23%, 0.36%, and 0.27%. The combination of these two operations further boosted accuracy. This demonstrates that the SEAS module effectively utilizes neighborhood spectral information to dynamically enhance the spectral features of the target pixels, further improving classification performance.

4) Effects of Key Hyperparameters on Model Performance: We performed a detailed analysis of key parameters, including learning rate, batch size, and the number of PCA components, to evaluate their influence on model performance. The results shown in Fig. 7 indicate that as the learning rate decreases, model performance improves across all three datasets, but it drops sharply when the rate falls below 0.0005, likely due to inadequate convergence. In contrast, batch size has minimal impact on performance. The number of PCA components, however, plays a pivotal role: for the IP dataset, increased components result in higher accuracy; for the HU dataset, optimal performance is achieved with 20–30 components, after which accuracy declines; and for the WHHH dataset, stable accuracy is achieved with 30–90 components, but further increasing the number of components leads to a decline in performance.

5) Effect of the Number of Training Samples: In Fig. 8, we evaluate the model performance under different training sample sizes. Fig. 8(a)–(c) presents a comparative analysis of our method and other approaches on the IP, HU, and WHHH datasets using sample sizes of 5, 10, 20, 30, 40, and 50 for each category. The results indicate a significant improvement in OA as the number of training samples increases across the three datasets, with our proposed method consistently outperforming the other comparison methods at all sample sizes. Notably, on the WHHH dataset, our method demonstrates superior classification accuracy even with 50 samples per category. This highlights that our proposed method not only achieves excellent classification accuracy with low sample sizes but also exhibits strong scalability.

6) Convergence Speed Analysis: The convergence speed and training stability of a model are key metrics for evaluating its performance. In Fig. 9, we further demonstrate the training loss curves of the HyperMamba model and other comparison methods on three datasets: IP, HU, and WHHH. It can be observed that the training loss curves of the proposed HyperMamba show a smooth decrease on all three datasets without significant oscillations. This indicates excellent training stability of the proposed HyperMamba model. While the convergence speed on the WHHH dataset is slightly slower compared to the IP and HU datasets, this is primarily due to the WHHH dataset containing more classes and being more challenging. When the number of epochs reaches 50, the training losses of HyperMamba are close to zero on all three datasets, whereas SSIN, SSGRN, DSGSF, and SS1DSwin require additional epochs to achieve convergence. Among all comparison methods, HyperMamba demonstrates strong training stability and convergence speed.

F. Visualization Results

To offer a more intuitive representation of our approach, in Figs. 10–12, we visualize the classification results of all compared methods on three datasets: IP, HU, and WHHH, respectively. Under the condition of ten training samples per

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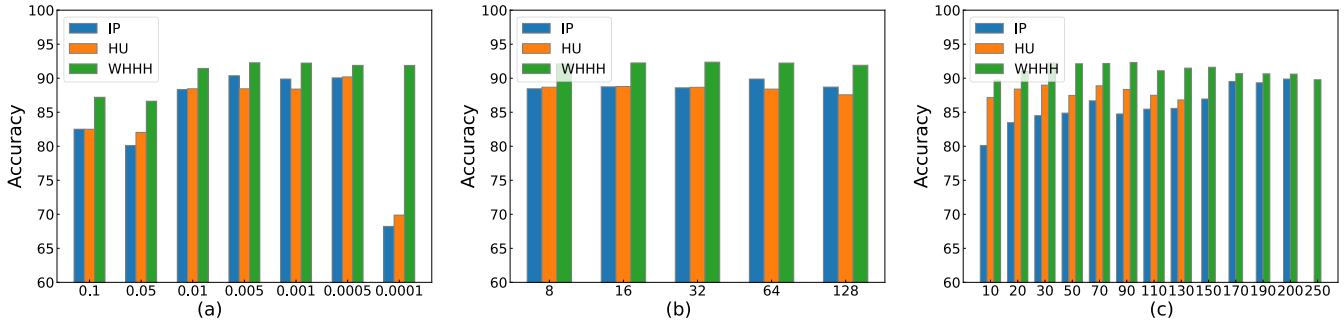


Fig. 7. Impact of key hyperparameters on OA across the IP, HU, and WHHH datasets. (a) Effect of learning rate. (b) Effect of batch size. (c) Effect of PCA components.

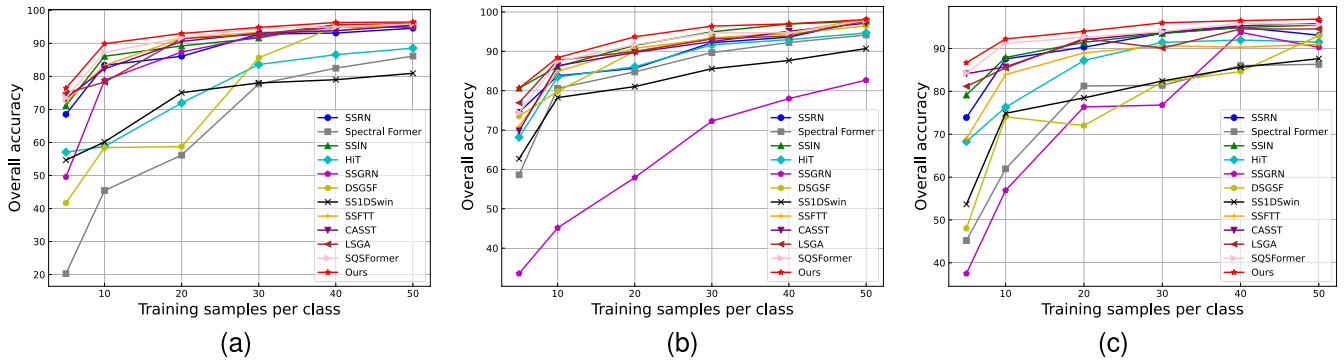


Fig. 8. OA across different training sample sizes. (a) IP dataset. (b) HU dataset. (c) WHHH dataset.

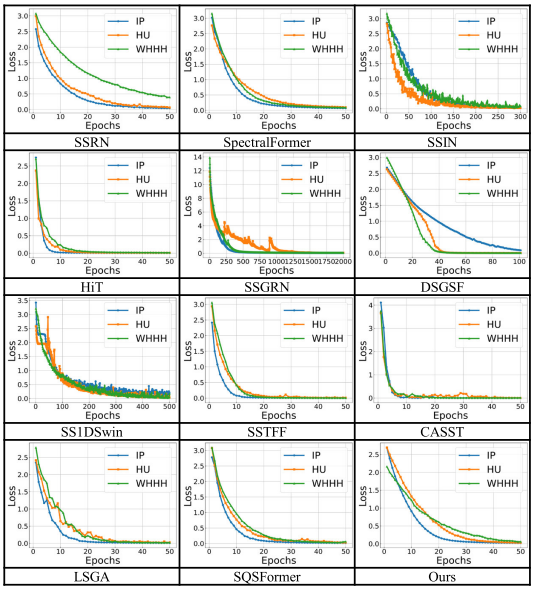


Fig. 9. Convergence curves of our method and comparison methods on the IP, HU, and WHHH datasets.

class, the performance of SpectralFormer and SS1DSwin is comparatively poorer. Conversely, other methods, especially SQSFormer and LSGA, demonstrate promising classification abilities. Our method, in comparison to these approaches, further reduces misclassified points. For instance, in the bottom-left region of the WHHH dataset, SQSFormer misclassifies "Cotto" as "Tree," while LSGA misclassifies "Red roof" as "Tree." In contrast, our method exhibits higher classification accuracy.

V. Discussion

In the quantitative analysis of Section IV, we thoroughly demonstrate the effectiveness of our method. In Section IV-C, Tables II–IV present the quantitative analysis results for the IP, HU, and WHHH datasets, indicating that our method achieves significant advantages. This evidence supports the claim that our proposed HyperMamba effectively enhances the interdependence between spatial and spectral information in high-dimensional HSIs, which contributes to improved classification performance. Notably, our method not only achieves the highest accuracy but also demonstrates strong performance in computational complexity, as shown in Table V. This validates that HyperMamba effectively captures spatial and spectral information while maintaining the computational efficiency of the Mamba architecture.

In addition, the ablation study in Table VII further confirms the effectiveness of the core modules, SNAS and SAES. SNAS adaptively selects the optimal neighborhood for spatial scanning by analyzing the spectral characteristics of classified pixels and balancing spatial neighborhood information with local spatial structure. The SAES module dynamically enhances the spectral features of classified pixels using neighborhood spectral information and conducts spectral scanning. The integration of the two modules further enhances both spatial and spectral information effectively. Figs. 8 and 9 further demonstrate that our method remains competitive under varying training sample conditions and highlight its advantage in training stability. The above experimental results show that this article provides a very promising method for analyzing high-dimensional HSIs.

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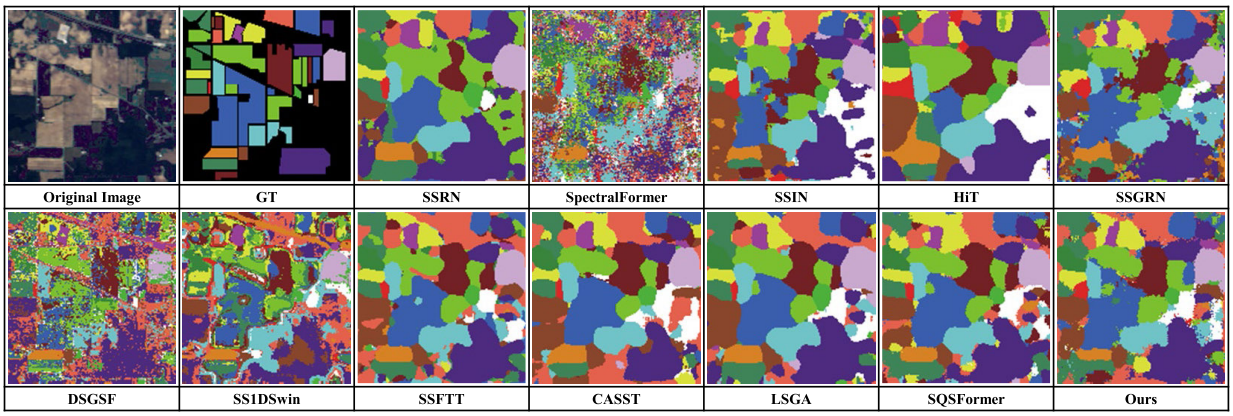


Fig. 10. Visualization of classification results on the IP dataset.

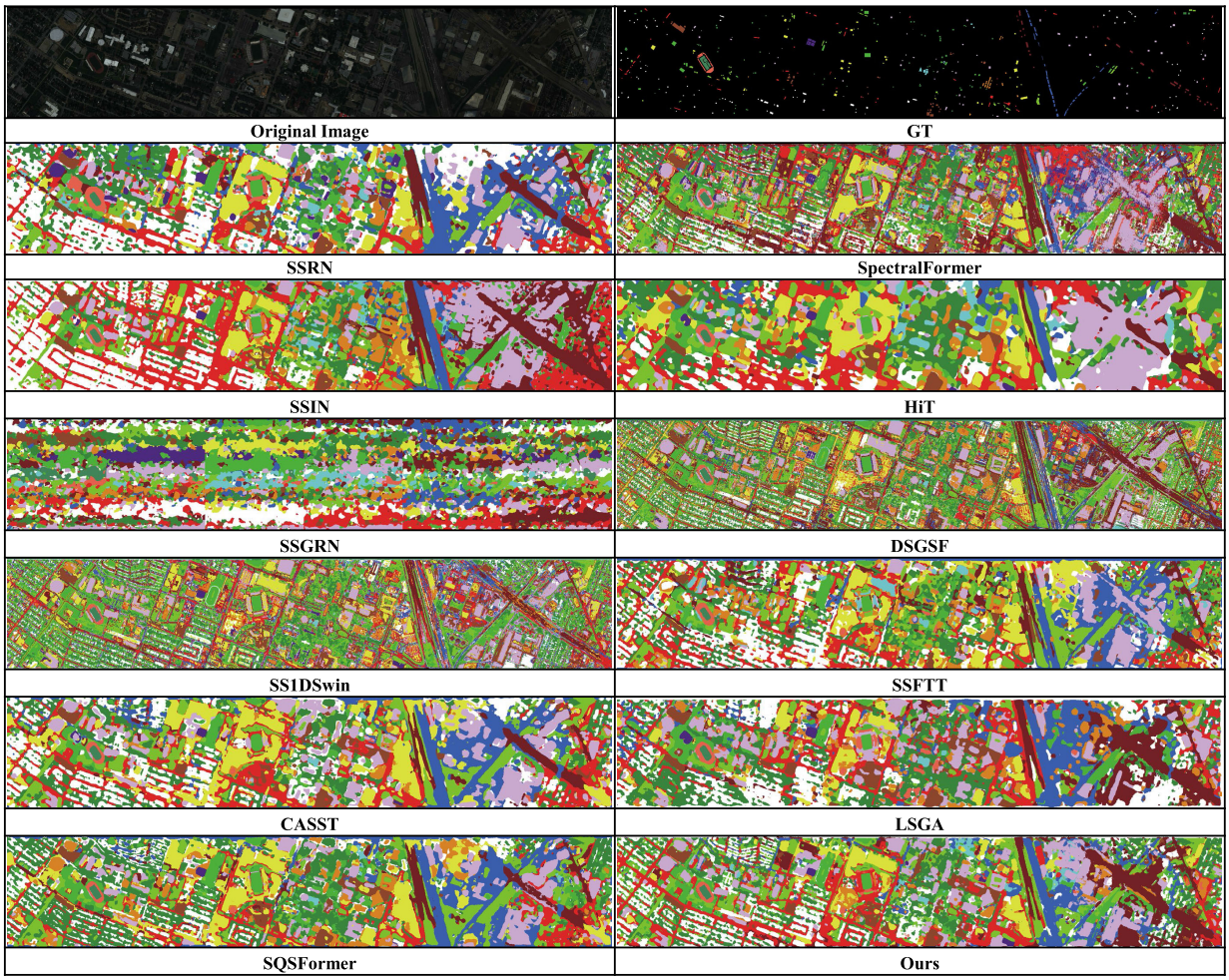


Fig. 11. Visualization of classification results on the HU dataset.

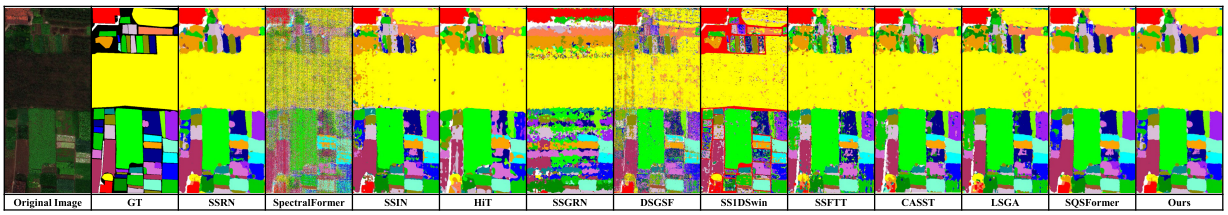


Fig. 12. Visualization of classification results on the WHHH dataset.

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VI. Conclusion

In this work, we proposed HyperMamba, a novel spectral–spatial adaptive Mamba model designed specifically for HSI classification. By SNAS and SAES, HyperMamba addresses the computational efficiency challenges caused by the high dimensionality of HSIs and the quadratic computational complexity of Transformer architectures. Experimental results across three widely used HSI datasets demonstrate the effectiveness of HyperMamba, surpassing existing state-of-the-art methods in classification accuracy while also maintaining computational efficiency. In future work, we will explore applying HyperMamba to various tasks, including HSI denoising and anomaly detection, to enhance its versatility in this field.

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